The fact that meteorites containing about 100 times excess fission tracks also contain ^{129}Xe is evidence both for the existence of ^{244}Pu at the beginning of the collapse of the solar nebula, and for the rapid condensation of the meteorites from freshly produced new elements. This time has been calculated to about 80 My from the isotopic concentrations. By similar analysis we can go further back in the history of the universe and calculate the time between the formation of the elements in the galaxy and the condensation of the solar system; the result yields a time of 9.9×10^9 y (or 9.9 eons).

There are also some radiochemical evidence about the age of our Galaxy. Consider Figure 12.12, which shows a part of a nuclide chart; the zigzag arrow shows nuclides formed in the s-process, the lower right arrows nuclides formed by the r-process. ^{186}Os can only form in the s-process and ^{187}Re only in the r-process. It is likely that our solar system, and ^{187}Re, was formed from a supernova in the r-process. ^{187}Re decays ($t_{½}$ 46×10^9 y) to ^{187}Os, which also is formed from slow n-capture in ^{186}Os. The amount of ^{187}Os formed in the s-process can be calculated from estimated flux values and known cross sections (at 30 keV; 3.6×10^8 K). Thus the amount of ^{187}Os formed by decay of ^{187}Re can be calculated, and the time since the r-process determined. This rhenium-osmium clock gives us an age of 10.8 ± 2.2 eons from formation of the Galaxy until our solar system formed. If this is combined with the age of the solar system of 4.7 eons, the age of the galaxy becomes 15.5 ± 2.3 eons. Adding 1.5 eons for the time between the Big Bang and the formation of the Galaxy, the universe should have an age of 17 ± 3 eons.

This value should be compared with the most recent value of t_o (12.5), which without a correction for any change in the rate of expansion, gives 13.18 ± 0.11 eons. The disagreement between the uncorrected value, 13.2 ± 0.1 eons, and the value, 17 ± 3 eons, above indicates either that the Universe does not continue to expand at an unchanged rate, or if it does, that

(i) the expansion of the universe accelerates slowly (as indicated by resent measurements), or
(ii) Hubble's law is not valid, or
(iii) the cosmological (Einstein) constant is not unity, or
(iv) something is wrong with the Big Bang hypothesis, or
(v) the radiometric data are too uncertain.

12.7. THE EVOLUTION OF THE PLANETS AND THE EARTH

The planets are assumed to have formed out of the same cloud as, and simultaneously with, the sun. The sun contains 99.85% of the mass of the solar system, the planet Jupiter 0.1%, and all other planets 0.04%. The originally spherical cloud was disturbed and caused to rotate through some external action (e.g. supernova explosion or passage of some celestial body), the rotation leading to the formation of a disk-like cloud, and later to cloudy rings similar to those one can see around many stars and some of the planets.

The cloud condensed successively into dust particles, which through collisions formed larger particles until rather large clumps of sizes similar to the present asteroids were obtained (so-called *planetesimals*, with diameters of several km; compare the sizes of the craters on the moon). By the time these clumps drifted together (a process referred to as *accretion*) to form the protoplanets, which collected more dust and gas from the surrounding cloud. When the H-burning started in the sun, a strong solar wind was produced, which blew away the gas and dust left in the cloud, and the planet formation was in principle finished. Some cosmogonists think this whole process took place in only a few million years.

The temperature gradient in the primordial elliptical cloud is given in Figure 12.13 with condensation temperatures for important elements and compounds. The volatile elements are almost completely absent in the innermost ("terrestrial") planets (Mercury, Venus, Earth, Mars). Here rock-forming minerals were produced in high-temperature chemical processes (Al_2O_3, SiO_2, $CaTiO_4$, $Ca_2Al_2SiO_7$, etc) and condensed out together with little volatile metals (Fe, Ni, etc). At lower temperatures further out in the solar system, H_2O was formed and condensed out as ice; e.g. some moons of Saturn are almost pure ice. At still lower temperatures NH_3 and CH_4 would liquefy. Jupiter and Saturn contain 99% H_2, 0.1% He, the rest being mainly H_2O (ice), CH_4 and NH_3; wide storms of NH_3-crystals have been observed on Saturn; Pluto has a thin atmosphere of only CH_4. Owing to the high pressures and low temperatures at the giant planets,

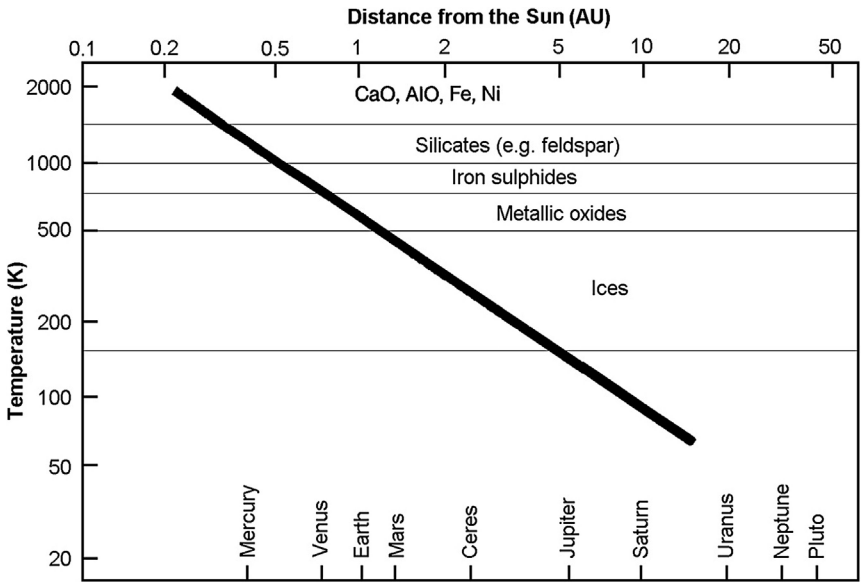


Figure 12.13 Temperature distribution in the solar system during planet formation. *(From Karttunen et al.)*

hydrogen is either in the liquid or solid (metallic) state. However, if the planets have migrated much over time to their present positions, this simple explanation does not hold.

From assumptions of the elemental composition and temperature-pressure conditions of the solar system one can in principle calculate the average mineral composition of the Earth using thermodynamic data of known compounds. Several such calculations have been made, which more or less well account for the present composition of the crust, magma, etc. There are, however, some other observations which need comments.

According to Urey, Suess, and others, the accretion of cosmic particles during the formation process released gravitational energy; also the higher concentration of radioactive nuclides in the primordial cloud (Fig. 12.14) contributed to heat and melt the solid material of our Earth-to-be. The Earth's temperature was 40−50% higher than today about 3.3 Gy ago. Not until a solid crust was formed could the radioactive clock begin to tick. This did not occur until about 3.7 eons ago, as this is the age of the oldest minerals found on Earth. Thus, it should have taken about 1 eon for the Earth to solidify. As soon as the temperature fell below 100°C water could begin to condense, possibly leading to an abrupt global temperature reduction, due to cloud formation and high heat of vaporization.

All planets were originally formed with some atmosphere, partly as a remnant of the primordial cloud, partly due to outgassing of the interior through volcanos and to impacts of comets. The atoms/molecules in the atmosphere have an average velocity, which must

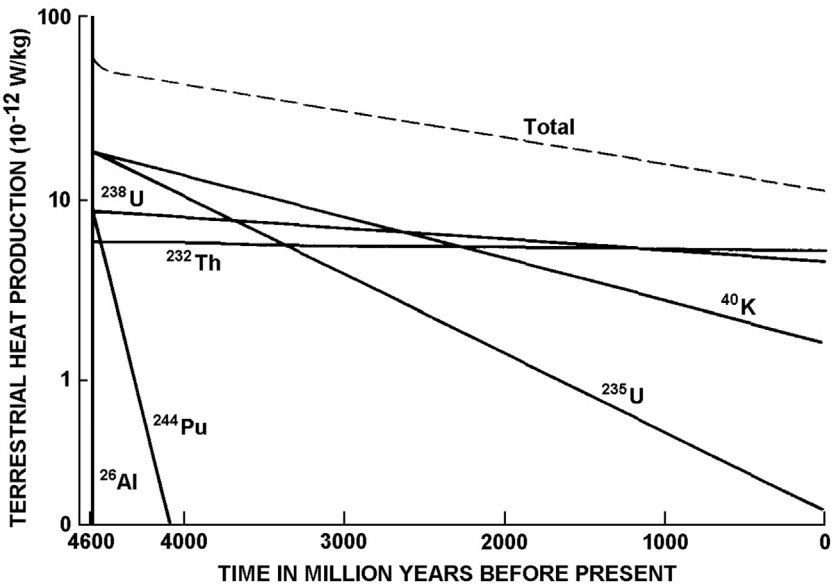


Figure 12.14 Heat production from decay of radioactive species in earth. *(From Brown and Musset, acc. to Cox.)*

be less than the escape velocity from the planet if the gases should not evaporate into space. The escape velocity depends of temperature and mass of the planet (cf. §12.1.2); for example, the moon has too small a mass to keep any atmosphere. The escape velocity of H_2 and He from the Earth is too large for these elements to be retained. The original atmosphere of the Earth is therefore believed to have been mainly H_2O, CO_2, N_2, CH_4 and NH_3. This is referred to as a *reducing atmosphere*, which probably is typical for all newborn Earth-sized planets. The UV-radiation from the sun decomposes these molecules (though the fragments may recombine), releasing H_2, which escapes. The O_2 released is very reactive and combines with many elements to form oxides; most of the red surface of Mars is FeO. The common assumption is that very little oxygen was originally present as free O_2. However, rather rapidly, processes led to a shift of the atmosphere from reducing towards *oxidizing conditions*. Life appears to have begun before this transition as evidenced by fossile bacteria-like organisms in very old rocks formed in a reducing atmosphere.

12.8. EXERCISES

12.1. (a) What is the most probable kinetic energy of a hydrogen atom at the interior of the sun ($T = 1.5 \times 10^7$ K)? (b) What fraction of the particles would have energies in excess of 100 keV?

12.2. What amount (kg) of hydrogen is consumed per second by the sun in the fusion reaction 12.6 in Table 12.2? The solar constant (energy flux at earth's orbit) is 1.353 kW m^{-2} and the earth's average distance to the center of the sun is 149 599 000 km.

12.3. Assuming the sun generates its energy to 2/3 by the proton-chain and the rest by the CNO cycle, what would the expected solar neutrino flux be at the center of the earth?

12.4. In a water power station with a fall height of 20 m and a water flow of 500 $m^3\ s^{-1}$, the electric power output at 100% efficiency can be calculated. If the heavy water in the fall could be extracted and converted in a fusion reactor according to (12.14) with a 25% efficiency, which of the two power sources would yield more energy?

12.5. The supernova SN1987A was at a distance of about 51.47 kPc from us and had about 10^{46} J total energy release as estimated from the neutrino flux hitting Earth. How much mass in kg was then converted to energy in the explosion?

12.9. LITERATURE

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